

No. of Functional Features Included in the Solution

Date	26 June 2026
Team ID	N/A
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Audio Upload	Upload audio file for analysis	Streamlit UI	Done	High
2	Audio Playback	Play uploaded audio	Streamlit	Done	Medium
3	Speech-to-Text	Convert speech into text	Whisper	Done	High
4	Semantic Similarity	Compare transcript with reference concept	NLP Module	Done	High
5	Audio Feature Extraction	Extract pause, energy, duration and speech features	Librosa	Done	High
6	Understanding Level Prediction	Predict Strong/Moderate/Poor understanding	Scoring Engine	Done	High
7	PDF Report Generation	Generate downloadable evaluation report	Report Module	Done	Medium
8	Visualization Dashboard	Display waveform and analysis results	Streamlit	Done	Medium

Feature Summary:

Metric	Count / Value
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Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	Speech-to-Text, Semantic Similarity, Audio Feature Extraction, Understanding Score

Additional / Nice-to-Have Features	PDF Report Generation, Audio Playback, Waveform Visualization
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Audio Upload, Audio Playback, Dashboard
2	Backend / Logic	3	Speech-to-Text, Semantic Similarity, Understanding Prediction
3	Database / Storage	1	Local File Storage for Audio and Reports
4	API / Integration	2	Whisper Model, Sentence Transformer
5	Security / Authentication	0	Not Implemented